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Numerical Simulation of Flow Over Buildings using OpenFOAM®.

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Abstract. The present work is a numerical simulation of flow over buildings. The numerical simulation was carried out using an open source Computational Fluid Dynamics (CFD) tool called OpenFOAM®. The Reynolds Averaged Navier Stokes Equations (RANS) was solved along with a two-equation turbulent closure. The solutions from the numerical simulations are post-processed with open source software namely Paraview which is based on Visualisation Tool Kit (VTK). Effect of flow over buildings is described in terms of contours of velocity. Further, the same data is processed with Line Integral Convolution (LIC) to reveal three-dimensional flow properties on a plane parallel to the inlet flow. The streamlines are visualised in addition to the LIC and velocity contours. The post processed results reveal the real flow physics with just LIC alone or with the help of streamlines and contours.

INTRODUCTION

Wind pattern around the tall buildings or tall structures may generate contrary and sometimes risky residential, industrial and environmental problem at an unimaginative level. This problem was observed and is in existence for nearly 2 to 3 centuries around the globe. This became the reason for the initial investigations using wind tunnel. Now a days Computational Fluid Dynamics (CFD) is being a very useful tool to simulate flow over an object, building and even a topography. Researches are being carried out using this tool.

Veena et al [1] determined the best possible location for installing wind turbines based on computational analysis. They used OpenFOAM® which uses pressure-implicit splitting operator (PISO) algorithm to solve the governing Navier Stokes equations by iterations. They were able to conclude that for a unit tall building, a significant component of winds is available to be used for generating wind power. For twin buildings, the placement of wind turbines can not only be at the roof top but also in between the buildings for using the tunnelling effect.

M. Ricci et al [2] studied the capabilities of Large Eddy Simulation (LES) which is a fluid flow analysis tool. As flow over a bluff body is difficult to analyse LES was used. Turbulent flow was simulated around the isolated buildings. Pressure field was determined and was compared with experimental results. Computational solution using domain with dimensions $L \times H \times W$ and experimental results were obtained by conducting experiments in ABL wind tunnel. The

convergence of simulation was checked for every angle of attack by analysing the pressure values. Vortical structures for 0,45,90 degrees were visualized qualitatively.

Pressure distribution and overall pressure distribution were also obtained. Z. Harun et al [3] studied the LES for the flow over the Petronas Twin Towers, the highest skyscraper in Kuala Lumpur, Malaysia. The simulation revealed the structure of the flow over the building. They used OpenFOAM® to simulate the results. The pressure-velocity coupling was maintained by the PISO algorithm. It was found that two large couples of vortices were generated in the flow; the first in-between the two towers and the second, in the downstream. The results show that the internal couple of vortices are much stronger but of little turbulence intensities. The flow generates two regions with high vorticity activities; the first is in-between the two towers and the second at the downstream. The flow is slightly affected by the PTT above height $2H$. The couple of vortices in-between the two towers is much stronger than the one in the wake region. However, its turbulence level is lower. Both couple of vortices becomes weaker at heights above H due to the diameter reduction of the two towers. Turbulence is transported vertically upwards in between the two towers and downwards in their wake.

Tsubasa Okaze and Akashi Mochida [4] generated a time series of wind velocity and fluctuations of air temperature which was based on the Cholesky decomposition of the Reynolds stress tensor and turbulent heat flux. An orthogonal grid system was applied. The mean wind velocity and the temperature of air predicted by LES using the generated inflow turbulence were generally in good agreement with the experimental results.

Marco Felipe King et al [5] made a computational analysis using Ansys Fluent and also with OpenFOAM® software. They compared it with original experimental measurements of airflow in and around a full-scale cubical structure both in isolation and within an irregular asymmetrical array. They investigated simple full-scale urban-like geometries, where they isolated the effects on ventilation to understand it better. This study was an idealised study which considers only wind-driven effects, the irregular asymmetrical arrays and its coupling, with openable façade components at full-scale providing a better opportunity to investigate ventilation under real wind conditions.

Federico Flores et al studied about the development of a numerical fluid dynamics solver in OpenFOAM® [6] framework to simulate the internal ventilation consisting of an open pit that includes the effects of induced turbulence, stratification and buoyancy. Simulations using open foam and experiments by generating LES and DES was carried out. URANS and the Detached Eddy Simulation techniques revealed similar results for simple non-buoyant flow cases. In strongly buoyant flow cases, the capability of DES to solve induced turbulence was not reproduced by URANS techniques.

Rallaon Dadioti and Simon J. Rees [7] examined the performance of conventional RANS turbulence models and detached eddy simulation (DES) techniques implemented in the open source code OpenFOAM®. Simulations were made for the flow over a unit tall building and was compared with the experimental estimations from wind tunnel. Results obtained concludes that, DES is significantly more demanding than RANS calculations but offers improves robustness and accuracy for a variety of wind conditions.

Francisco TojaSilva et al [8] simulated the flow of wind around a building kept in isolation using various turbulence models under RANS using OpenFOAM®. The behaviour of the various RANS models was studied at various regions of the building roof like upstream side, downstream side and the mid region of the roof. The most appropriate turbulence models under RANS were identified for all the situations both qualitatively and quantitatively.

COMPUTATIONAL METHODOLOGY

To simulate the flow over buildings, an open source CFD software namely OpenFOAM® has been used. The present case is an example case available in OpenFOAM. The set of buildings over which the flow pattern is to be simulated is placed at the middle of a rectangular domain with z axis denoting the height of the buildings (refer to Figure 1). The streamline pattern around this model would be applicable to most of the cities as the model contains buildings with varying height and width. Short buildings are placed in front of tall buildings and vice versa which is a typical example for most of the urban area.

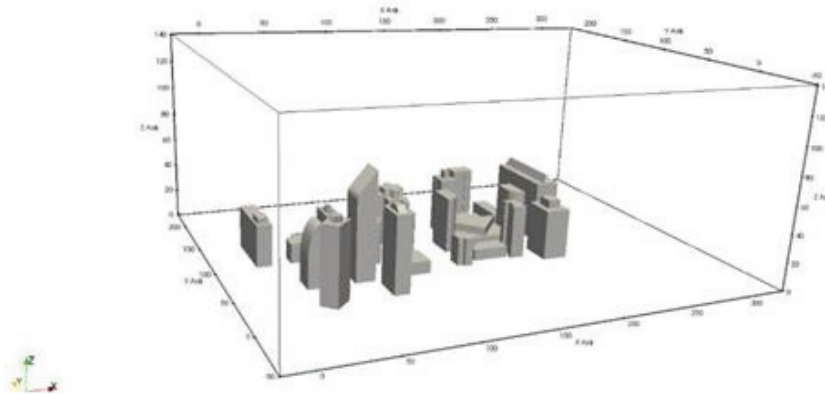


FIGURE 1. Building domain used for analysis in OpenFOAM®.

The computational domain is divided into smaller cells to form a mesh where the governing equations are solved. The figure 2 shows a plane passing from the inlet to the outlet along with the mesh. It is clear from the figure that the mesh is fine at the surface of the buildings and becomes coarse as we move away from the buildings. Meshing in such a way would result in better accuracy near the surface of the building.

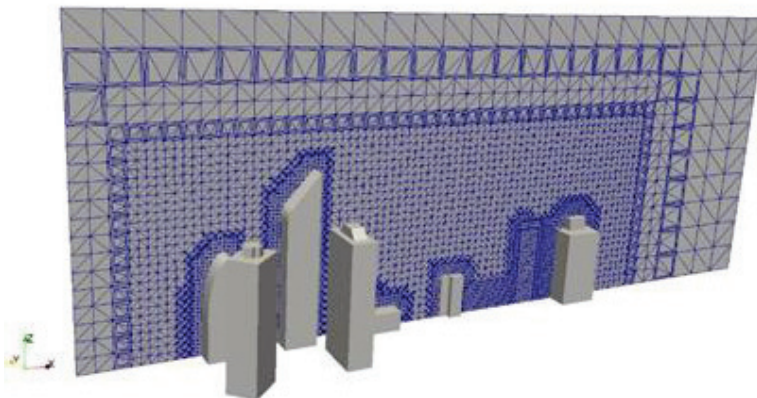


FIGURE 2. Quad type mesh showing grids coarser away from the building surface.

The governing equations consisting of Reynolds Averaged Navier Stokes equations along with the two-equation turbulence closure model namely standard k-epsilon model was used to solve via SIMPLE algorithm as implemented in OpenFOAM as simpleFoam. The inlet velocity is fixed at 10m/s with turbulent intensity of 0.1 based on which the k and epsilon was calculated. The kinematic viscosity was fixed at $1.5 \times 10^{-5} \text{ m}^2/\text{s}$. The results are post processed using Paraview and the same is discussed in the following section.

RESULTS AND DISCUSSION

The post processed results are presented in this section as contours, streamlines and LIC visualizations. The figure 3 shows the streamline pattern around the buildings subjected to the flow. The streamlines are injected from the line source place in between the buildings oriented along the Y-axis. The streamlines are integrated on both the directions using Runge-Kutta 4-5 order method. This is to ensure there is no missing of flow path or streamline pattern at some places. The resulting streamline is coloured by velocity magnitude to reveal the re-circulation zones and the normal flow zones.

The initial upstream flow is quite undisturbed and as the flow passes over the buildings, there is a production of three-dimensional wake which includes some stagnant flow and complex flow patterns. The streamlines show spiral behaviour near the downstream buildings and it also reveals the flow in lateral directions in comparison to the major flow being in the longitudinal direction.

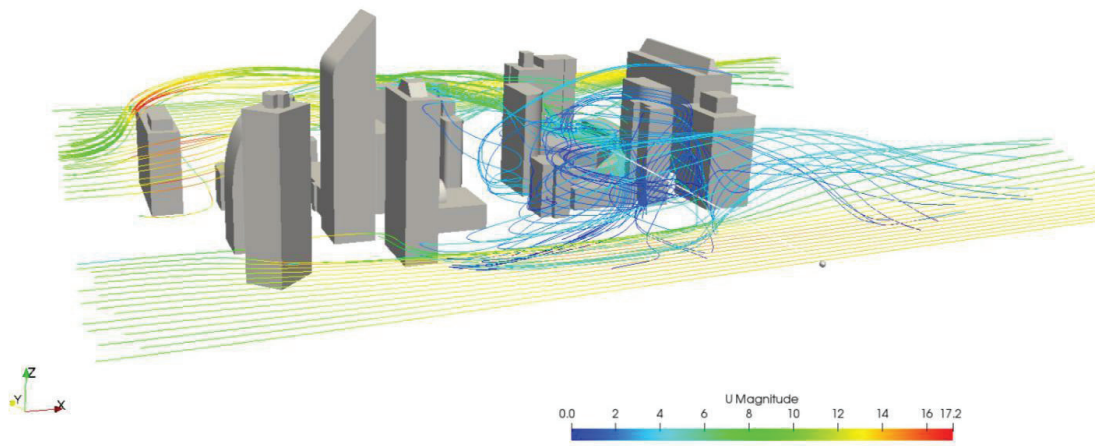


FIGURE 3. Streamline pattern around the building model.

The figure 4 shows the velocity contour in a vertical plane that extends from the inlet to the outlet. The contours reveal the acceleration of flow over the buildings especially the curved ones. The top of few buildings acts like a convergent section and it accelerates the flow to show higher values of velocity magnitudes. The lower values of velocity magnitude indicate stagnant zone or a re-circulation zone and it is seen usually behind the buildings. The wake zone is also seen clearly by the contours of velocity magnitude.

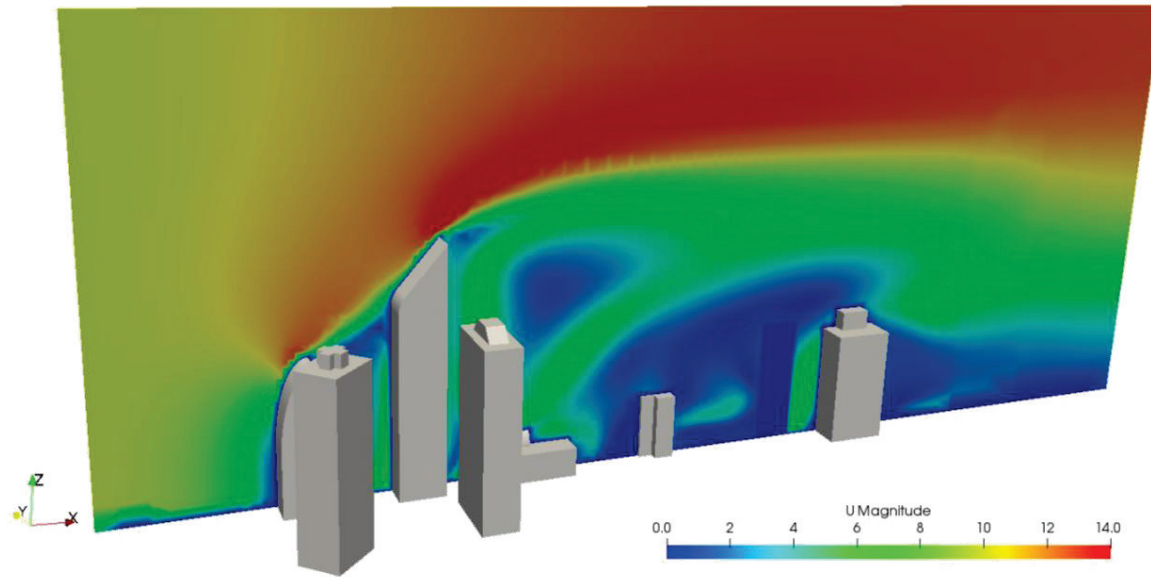


Figure 4. Velocity contours on a vertical plane passing through building

The figure 5 shows the velocity contour with Line Integral Convolution (LIC) visualization. The LIC contours are multiplied with the velocity magnitude and the contours are enhanced along with proper selection of parameters to create better results. The contours show the traces of velocity magnitude in terms of colour and also looks like streamlines drawn on the plane. The contours show flow structures both in plane and out of plane.

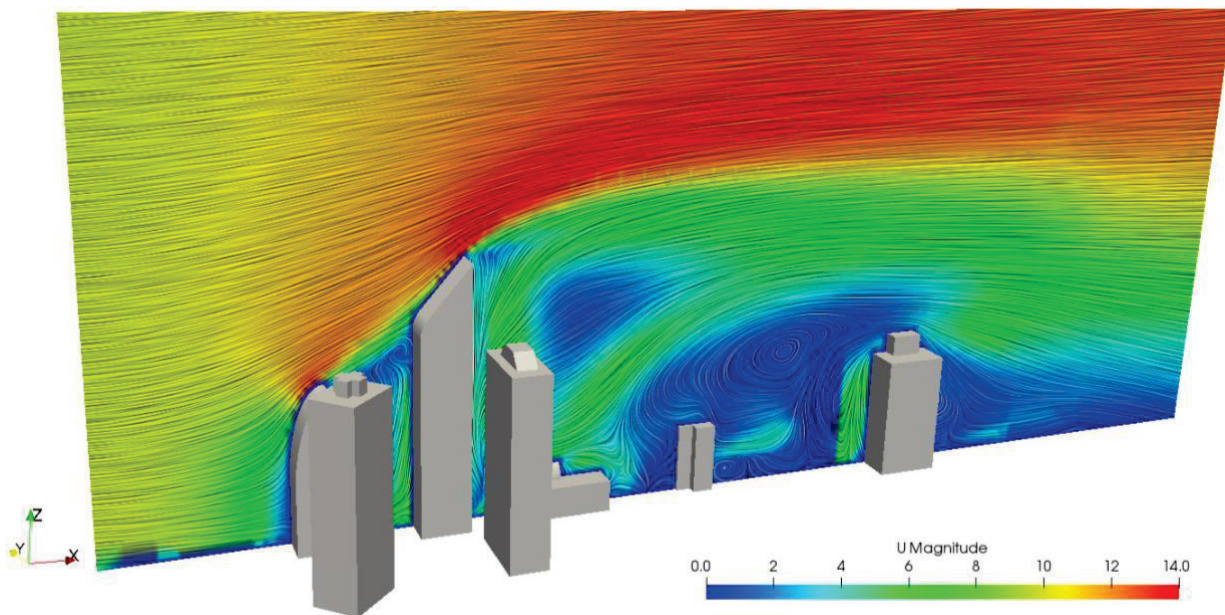


Figure 5. Velocity contour plot showing LIC visualisation

The figure 6 shows LIC visualization on a plane passing through buildings at the farthest end of y-axis in the positive direction. The flow features revealed by the LIC contours include rotational parts, in plane and out of plane motions. This visualization also reveals some physics like nodal points, sources, sinks, lines of merger, injections and ejections and the same shall be explored both experimentally and mathematically to get deeper insights. LIC is found to be a better way of post processing visualization in complex flow fields like flow between buildings.

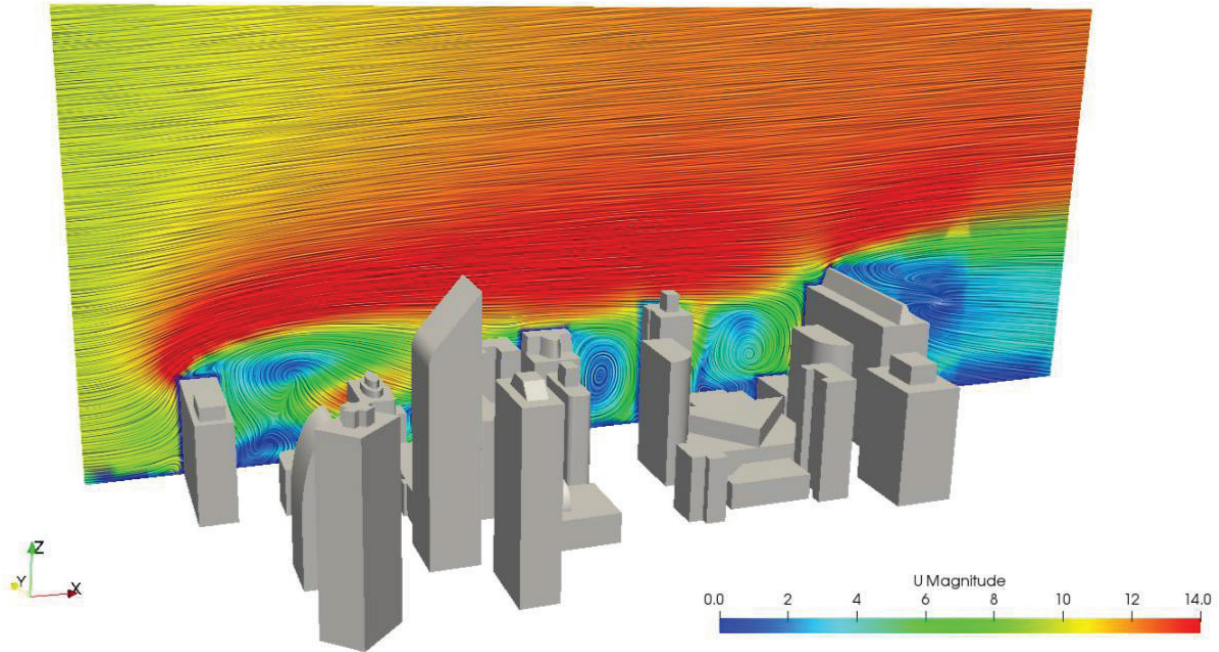


Figure 6. LIC visualization in positive end of y-axis

CONCLUSIONS

The present work deals with the numerical simulation of flow over buildings using an open source OpenFOAM®. A rectangular computational domain was used to study the effect of flow over the buildings. The contours of velocity only revealed the magnitude and streamlines provided the three-dimensional data. Further, the same data is processed with Line Integral Convolution (LIC) to reveal three-dimensional flow physics on a plane parallel to the inlet flow. The streamlines are visualised and the same is found difficult to analyse the flow physics. For complex flows such as flow over buildings, LIC was found advantageous over the conventional post processing methods.

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